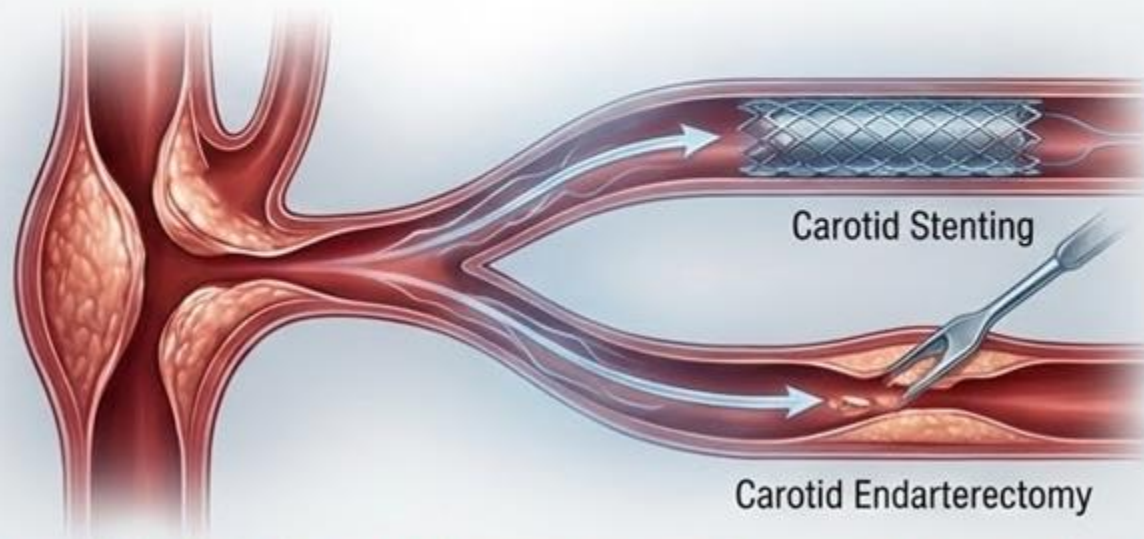
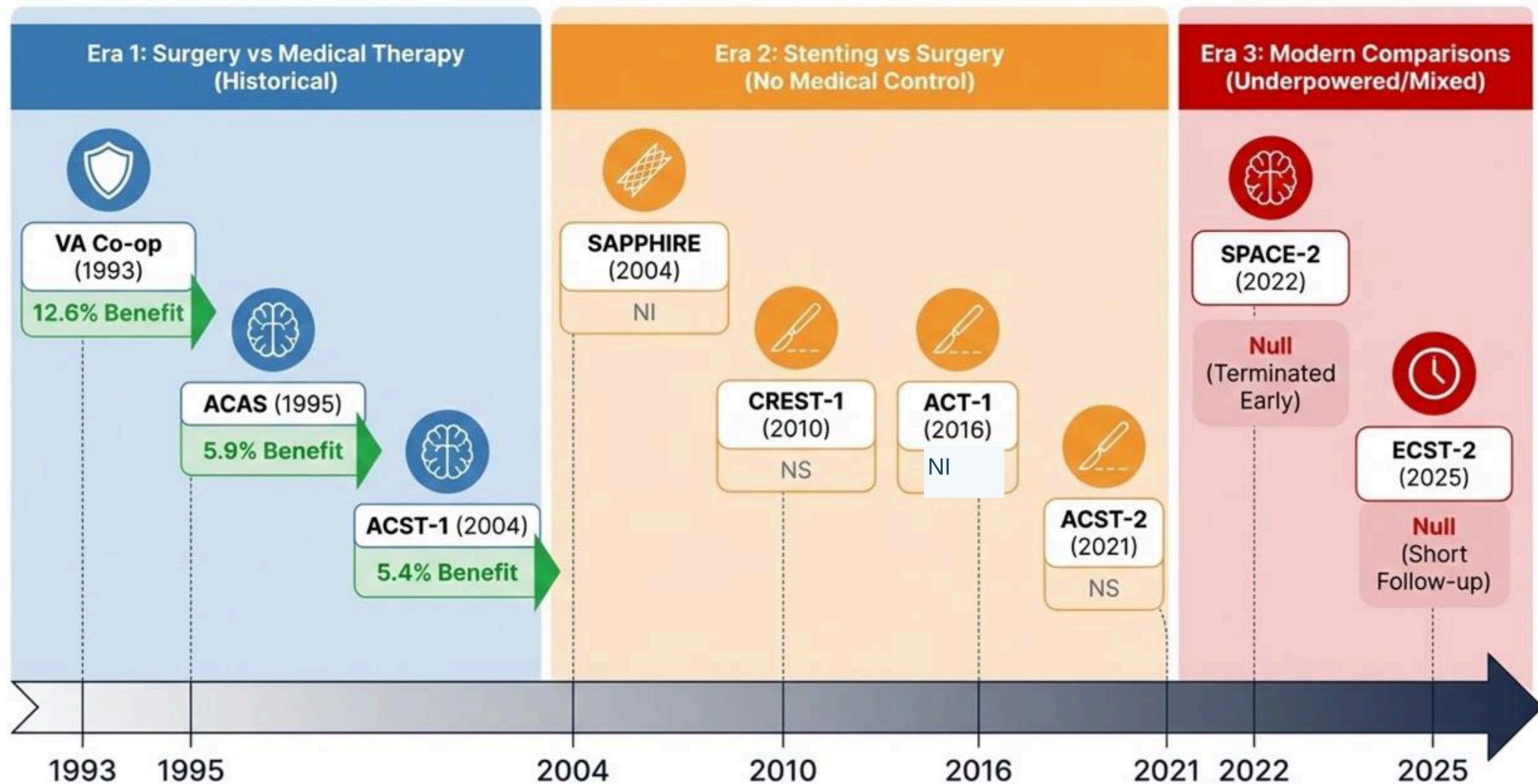


Medical Management and Revascularization for Asymptomatic Carotid Stenosis



Brott TG, Howard G, Lal BK, et al. Medical Management and Revascularization for Asymptomatic Carotid Stenosis. *N Engl J Med*.



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This timeline is historical and not a standardized comparison of benefits. ACT I established noninferiority within a prespecified margin; a nonsignificant comparison is not proof of equivalence. Trials used different endpoints, procedural windows and medical treatment.

pubmed.ncbi.nlm.nih.gov/7723155/
pubmed.ncbi.nlm.nih.gov/26886419/
pubmed.ncbi.nlm.nih.gov/36115360/
pubmed.ncbi.nlm.nih.gov/15470212/
pubmed.ncbi.nlm.nih.gov/20505173/

Trial (Year)	Population & Stenosis	Comparison	Primary Outcome (ARD)	Safety (30d S/D)	
Era 1: Surgery vs Medical Therapy (Historical)					
VA Co-op (1993)	Asympt Men >50%	CEA vs Med	Ipsilateral TIA/Stroke 8.0% vs 20.6% (12.6% Benefit)	4.7% (CEA)	Men only; TIA outcome included
ACAS (1995)	Asympt >60%	CEA vs Med	5y Ipsi Stroke/Death 5.1% vs 11.0% (5.9% Benefit)	2.3% (CEA)	Aspirin
ACST-1 (2004)	Asympt >60%	Immediate vs Deferred CEA	5y Any Stroke/Death 6.4% vs 11.8% (5.4% Benefit)	3.1% (CEA)	Variable statin use
Era 2: Stenting vs Surgery (No Medical Control)					
SAPPHIRE (2004)	Mixed (High Risk) Asym>80%	CAS vs CEA	1y D/S/MI 12.2% vs 20.1% (NI)	4.8% vs 5.4%	High risk cohort only; MI drove the difference
CREST-1 (2010)	Mixed Asymp >70%	CAS vs CEA	4y D/S/MI 7.2% vs 6.8% (NS)		Mixed (53% Symptomatic)
ACT-1 (2016)	Asympt >70%	CAS vs CEA		2.9% vs 1.7%	CAS perioperative-stroke ~2x CEA
ACST-2 (2021)	Asympt >60%	CAS vs CEA			
Era 3: Modern Comparisons (Underpowered/Mixed)					
SPACE-2 (2022)		Revasc vs Med			Terminated early (n=513); Underpowered
ECST-2 (2025)	Mixed >50%	Revasc vs Med			Mixed population; Short follow-up

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ACAS medical care included aspirin and risk-factor management. CREST-1 safety rates 2.5% versus 1.4% refer to its asymptomatic subgroup. ACT I met noninferiority. ACST-2: 5-year nonprocedural stroke 5.3% versus 4.5%; procedural stroke/death 3.7% versus 2.7% among patients who underwent the procedure. ECST-2 primary hierarchical win ratio 1.01 (95% CI 0.60-1.70); its 10.5%/10.2% pair was secondary, and the unsupported 1.2% safety estimate is removed. SPACE-2 used an ECST stenosis threshold and a perioperative/later ipsilateral composite.

pubmed.ncbi.nlm.nih.gov/34469763/
pmc.ncbi.nlm.nih.gov/articles/PMC8473558/
pubmed.ncbi.nlm.nih.gov/40252662/
pubmed.ncbi.nlm.nih.gov/7723155/
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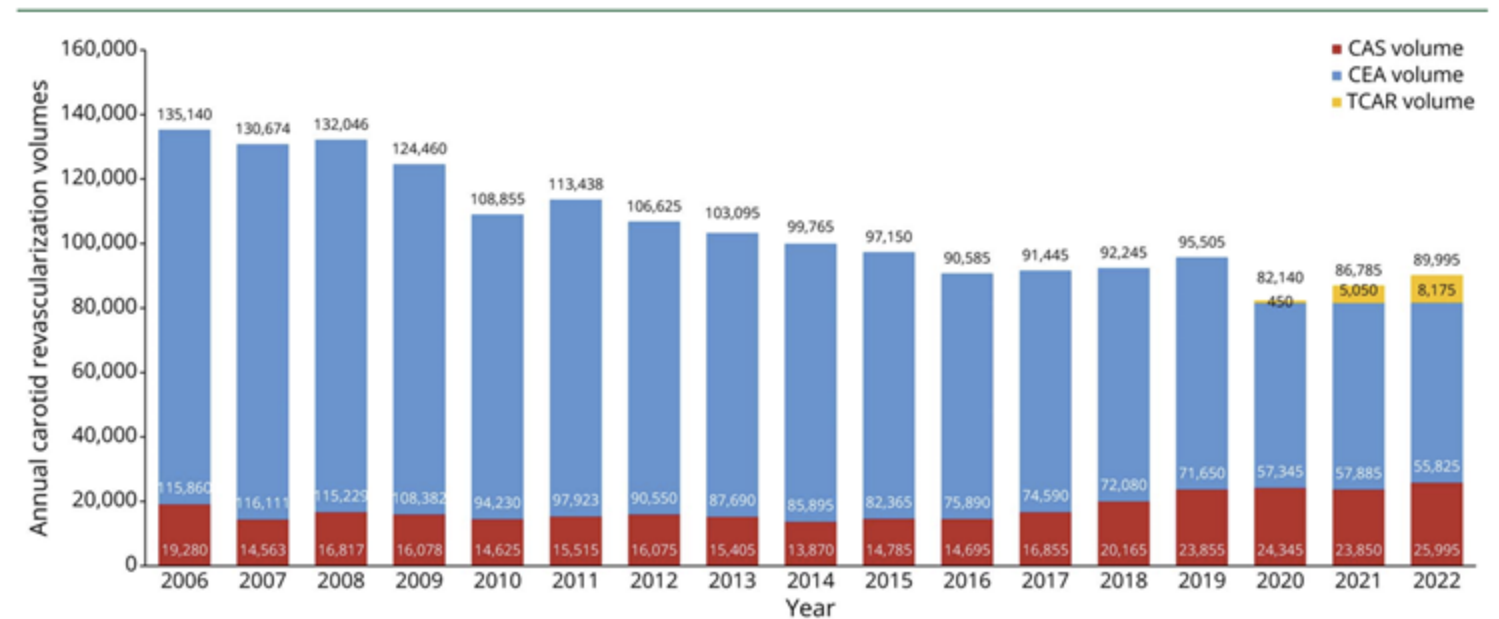
Trends in Carotid Revascularization: United States

Utilization rates (2006-2022)

- *Overall*: ↓ -3.9%/yr until 2015
- *CEA*: ↓ -5.5%/yr
- *CAS*: ↑ +12.0%/yr since 2016
- *TCAR*



Figure 1 Annual Volumes of Carotid Revascularizations in the United States From 2006 to 2022 According to Revascularization Type



TCAR estimates are for the period October 2020–December 2023. CAS = carotid artery stenting; CEA = carotid endarterectomy; TCAR = transcrotid artery revascularization.

Neurology. 2025;105(8):e214104.

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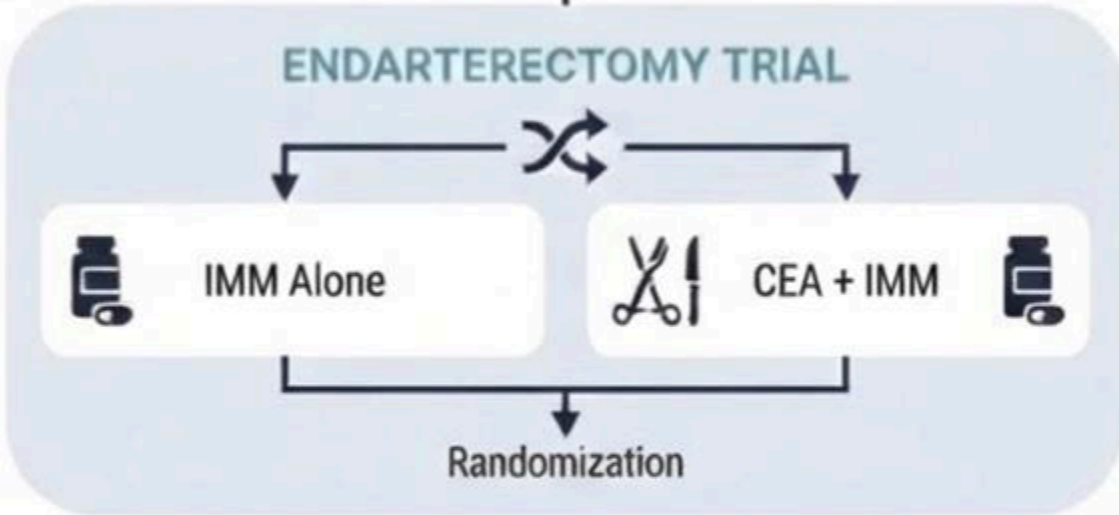
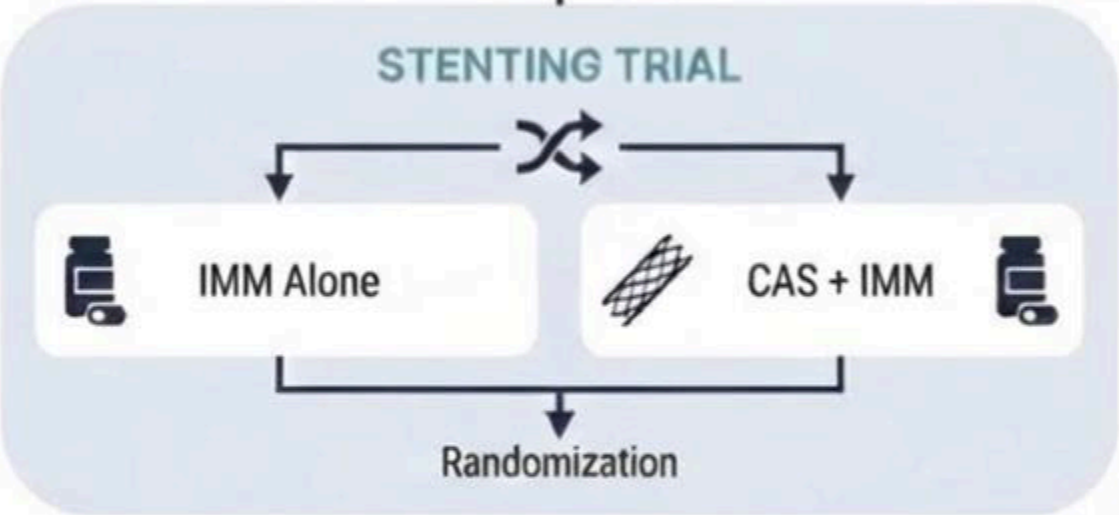
The annual percentage changes describe standardized utilization per 100,000 population, not raw procedure volumes. The asymptomatic proportion derives from four state databases with differing coverage periods.

pubmed.ncbi.nlm.nih.gov/40971741/

CREST-2



2 Parallel Trials



ENROLLMENT

- 155 Centers
- 5 Countries

OPERATOR REQUIREMENTS

- Experienced
- Periprocedural Stroke & Death Rate <3%

PATIENT ELIGIBILITY

- Age ≥ 35 years
- Stenosis $\geq 70\%$ (Ultrasound, CTA, MRA, DSA)
- No Stroke, TIA, or Retinal Ischemia ≤ 6 months

EXCLUSIONS

- Prior Disabling Stroke
- Unstable Angina
-

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The AF exclusion was atrial fibrillation prompting anticoagulation, not every AF diagnosis. Interpret the eligibility diagram with the complete original criteria.

pmc.ncbi.nlm.nih.gov/articles/PMC2932446/
pmc.ncbi.nlm.nih.gov/articles/PMC13148431/

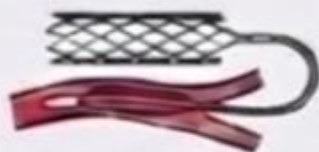
Treatment Procedures: Intensive Medical Management and Revascularization



Intensive Medical Management (IMM)

- **Protocol:** Identical across all arms (except periprocedural antiplatelets)
- **Primary Targets:** SBP < 130 mmHg, LDL < 70 mg/dL
- **Management:** Glucose, HbA1c, lifestyle factors (smoking, weight, activity)
- **Telephone health coaching**

Procedure



Procedure: Local anesthesia, embolic protection required



Carotid Endarterectomy (CEA)

Pre-procedure Antiplatelets (≥48h before)

Aspirin (325 mg/d) + Clopidogrel (75 mg bid)

Aspirin (325 mg/d)

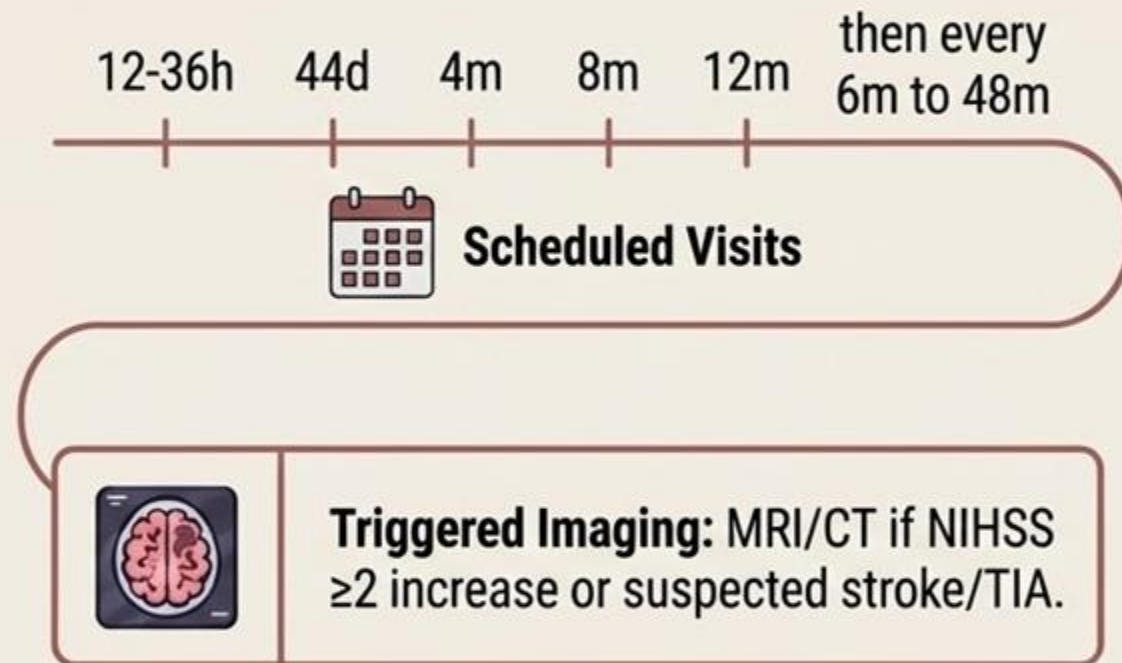
Post-procedure Antiplatelets

Clopidogrel (75 mg/d) + Aspirin (75-325 mg/d) for 30d, then Aspirin (70-325 mg/d)

Aspirin (70-325 mg/d)

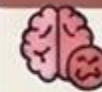
Follow-up Assessments and Outcomes

Follow-up Assessments



Primary and Secondary Outcomes

Primary Outcome (4-Year Composite, ITT)



Periprocedural ($\leq 44d$): Any Stroke or Death



Postprocedural ($>44d - 4y$): Ipsilateral Ischemic Stroke

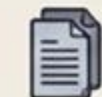
Stroke Definition & Adjudication



Blinded Committee, WHO criteria



Classification: Major (NIHSS ≥ 6), Disabling (mRS ≥ 3 @ 30d), Ischemic/Hemorrhagic.



Secondary Outcomes: Primary Outcome + Postprocedural Contralateral Stroke, Tissue-Based Infarction / ICH / SAH.

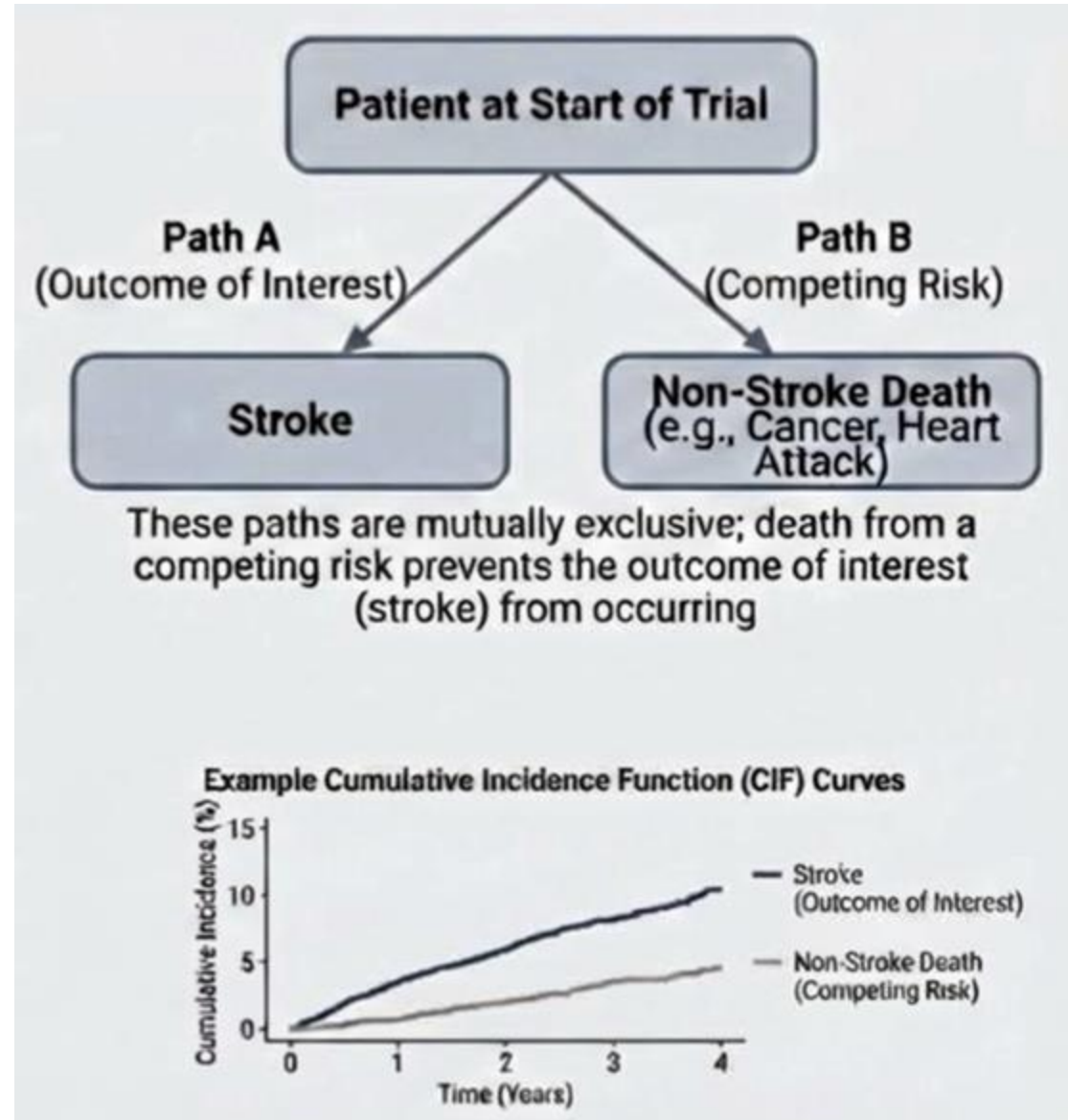
Source-qualified teaching update | 30 September 2026

The WHO symptom-duration boundary is at least 24 hours. Secondary tissue-based outcomes were defined composites including specified infarction/hemorrhage and death components, with territory and time-window limits; they are not a count of routine MRI lesions alone.

pubmed.ncbi.nlm.nih.gov/41269206/
doi.org/10.1056/NEJMoa2508800

Analysis

- 1° analysis (4-year ITT): Kaplan-Meier & CIF (accounting for competing risk of death)
- Power: 85%, 3.6% 4-year event rate in CAS/CEA groups, $\alpha = 0.05 \rightarrow$ detect medical tx rates $<0.8\%$ or $>8.4\%$
 - 5% crossover & 2.5% annual withdrawal
- Estimation: 95% CIs via bootstrap methods (100K reps)
- No correction for multiplicity (2° outcomes) or imputation (missing data)



Results

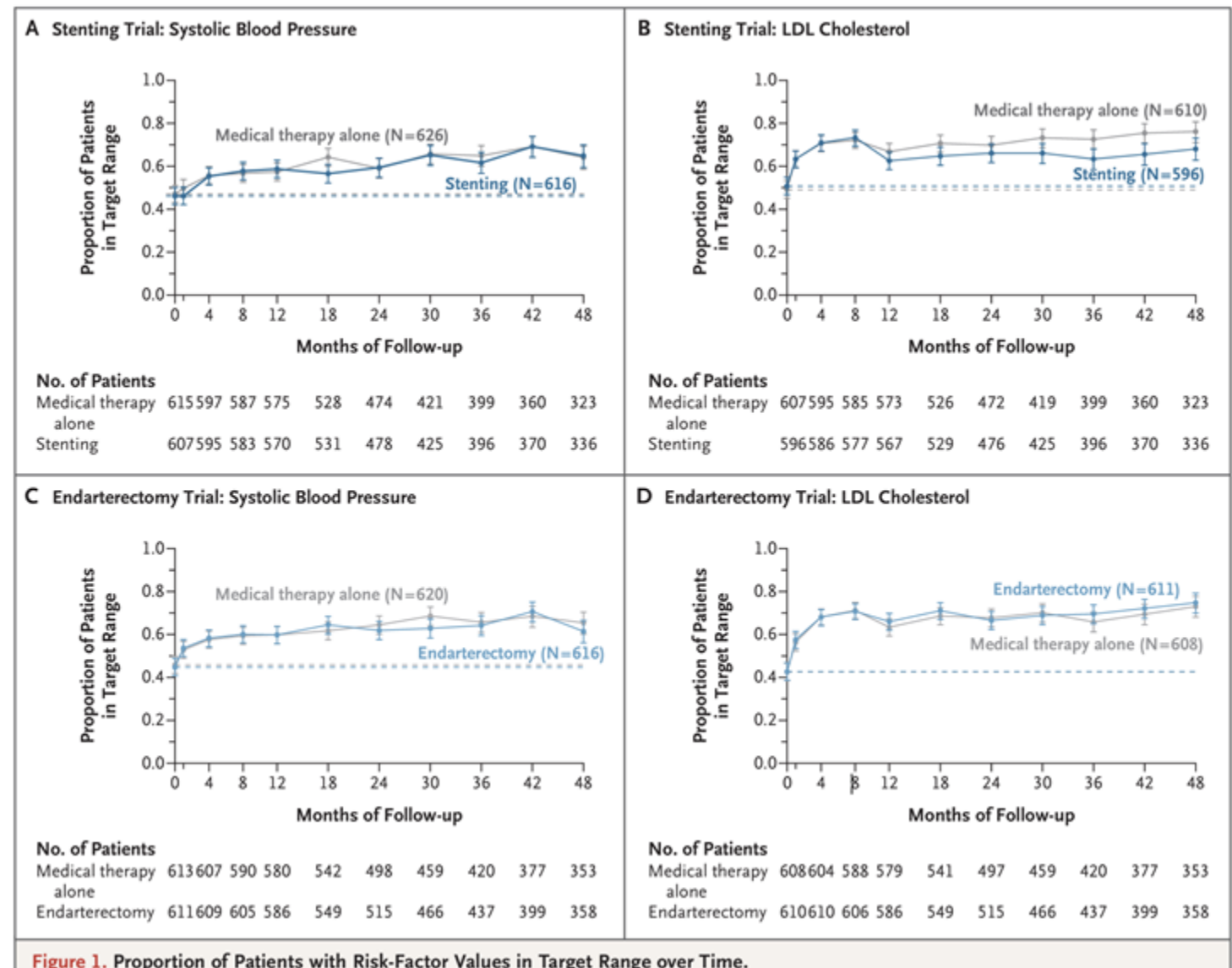
- mean age ~70 yr
- ~ $\frac{2}{3}$ men
- ~90% White
- ↑ vascular RF burden
- ~90% mRS=0

Table 1. Demographic and Clinical Characteristics of the Patients in the Two Trials, According to Treatment Assignment.*

Characteristic	Stenting Trial (N = 1245)		Endarterectomy Trial (N = 1240)	
	Medical Therapy Alone (N = 629)	Stenting (N = 616)	Medical Therapy Alone (N = 623)	Endarterectomy (N = 617)
Age — yr	69.7±7.7	69.3±8.1	70.4±7.6	70.7±7.8
Female sex — %	38.2	36.9	39.0	35.3
Race — %†				
White	90.1	92.9	88.3	90.0
Black	6.2	5.7	6.9	6.0
Other, not reported, or missing data	3.7	1.5	4.8	4.1
Hispanic ethnic group — %†	4.7	4.8	3.9	4.4
Previous stroke or TIA on target lesion >180 days before randomization — %	4.9	8.0	8.4	8.9
Risk factors — %				
Hypertension	87.4	88.0	84.9	85.1
Diabetes	37.8	40.7	38.0	34.4
Dyslipidemia	93.3	92.0	90.0	91.5
Current smoking	21.0	18.8	21.2	21.1
Previous cardiovascular disease or CABG	54.5	53.7	43.3	44.3
Blood pressure — mm Hg				
Systolic	138.8±20.2	138.2±20.2	137.9±19.8	139.3±20.2
Diastolic	73.2±10.8	73.1±11.2	72.6±10.2	73.1±10.7
LDL cholesterol — mg/dl‡	76.7±34.6	77.1±36.5	81.3±33.9	80.3±33.6
Body-mass index§	28.7±5.6	29.3±5.5	28.5±5.4	28.7±5.3
Stenosis at randomization — %				
Index artery				
≥70% stenosis	97.6	97.7	97.1	97.4
Peak systolic velocity ≥389 cm/sec¶	33.5	31.1	32.9	37.2
Nonindex artery: ≥50% stenosis	34.4	37.0	37.2	35.7
Modified Rankin scale score of 0 — %	87.8	88.8	87.9	87.0
CHA ₂ DS ₂ -VASc score of ≥4 — %**	56.9	60.1	53.8	55.9

Medical Mgmt.

- ~70% achieved SBP & LDL-C targets



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Broadly similar measured risk-factor control does not establish absence of performance bias in an open-label trial. Masked endpoint adjudication is a separate design feature.

pubmed.ncbi.nlm.nih.gov/41269206/

doi.org/10.1056/NEJMoa2508800

CAS:

- NNT $\cong 31$
- periprocedural risk < long-term risk reduction

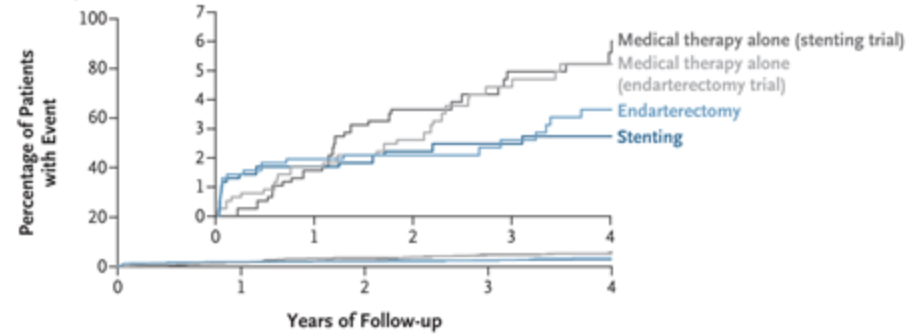
CEA:

- ARD 1.6%
- underpowered for small effect size

Table 2. Analysis of Primary Outcome and Components.

Variable	Stenting Trial		Endarterectomy Trial	
	Medical Therapy Alone	Stenting	Medical Therapy Alone	Endarterectomy
Primary 4-yr composite outcome*				
Event rate (95% CI) — %	6.0 (3.8 to 8.3)	2.8 (1.5 to 4.3)	5.3 (3.3 to 7.4)	3.7 (2.1 to 5.5)
Absolute difference (95% CI) — percentage points†	3.2 (0.6 to 5.9)		1.6 (–1.1 to 4.3)	
P value for difference	0.02		0.24	
Relative risk (95% CI)†	2.13 (1.15 to 4.39)		1.43 (0.78 to 2.72)	
Components of primary outcome				
Periprocedural period: stroke or death				
No. of events/no. of patients	0/629	8/616	3/623	9/617
Percent of patients with event (95% CI)	0.0 (0.0 to 0.6)	1.3 (0.6 to 2.5)	0.5 (0.1 to 1.4)	1.5 (0.7 to 2.8)
Difference (95% CI) — percentage points	–1.3 (–2.2 to 0.4)		–1.0 (–2.1 to 0.1)	
Postprocedural period: ipsilateral ischemic stroke				
No. of person-yr	1686	1714	1761	1823
No. of events/no. of patients	28/600	7/582	23/600	10/596
Annual event rate per person-yr (95% CI) — %	1.7 (1.1 to 2.4)	0.4 (0.2 to 0.9)	1.3 (0.9 to 2.0)	0.5 (0.3 to 1.0)
Relative risk (95% CI)	4.07 (1.78 to 9.31)		2.38 (1.13 to 5.00)	

A Primary-Outcome Analysis

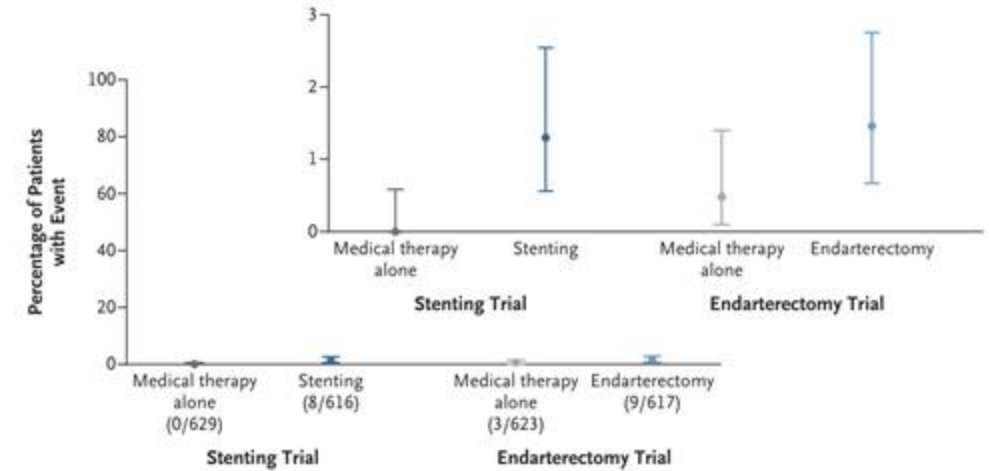


No. of Patients

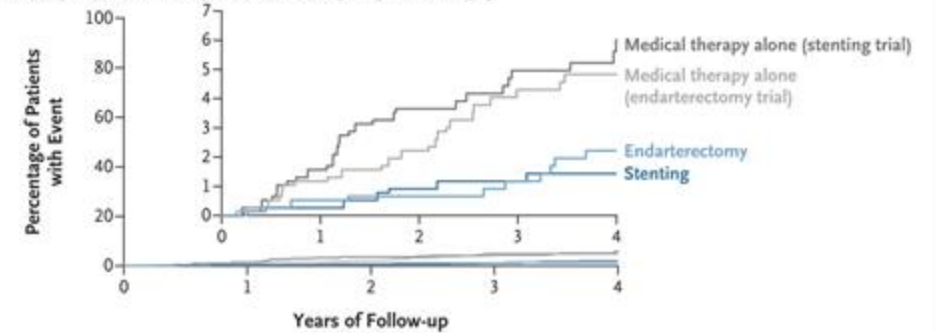
Stenting trial					
Medical therapy alone	629	537	428	360	276
Stenting	616	543	431	375	289
Endarterectomy trial					
Medical therapy alone	623	550	461	381	304
Endarterectomy	617	561	482	411	319

Metric	CAS Trial (Sig)	CEA Trial (NS)
Periprocedural Events (0-44d)	1.3% (7 strokes, 1 death)	1.5% (9 strokes)
IMM Arm Risk (0-44d)	0%	0.5%
Post-procedural Annual Stroke Rate	0.4% vs 1.7%	0.5% vs 1.3%
Safety Threshold (<3%)	Met (1.3%)	Met (1.5%)

B Periprocedural Component of Primary-Outcome Analysis (days 0 to 44)



C Postprocedural Component of Primary-Outcome Analysis (after 44 days)



No. of Patients

Stenting trial					
Medical therapy alone	600	537	428	360	276
Stenting	582	543	431	375	289
Endarterectomy trial					
Medical therapy alone	600	550	461	381	304
Endarterectomy	596	559	480	409	317

Secondary Outcomes

OUTCOME	CAS (VS MED)	CEA (VS MED)
Primary (WHO) Clinical sx >24h	2.8% vs 6.0% SIG (P=0.02)	3.7% vs 5.3% NS (P=0.24)

Disabling Stroke mRS ≥ 3	1.6% vs 1.2%	1.5% vs 0.5%
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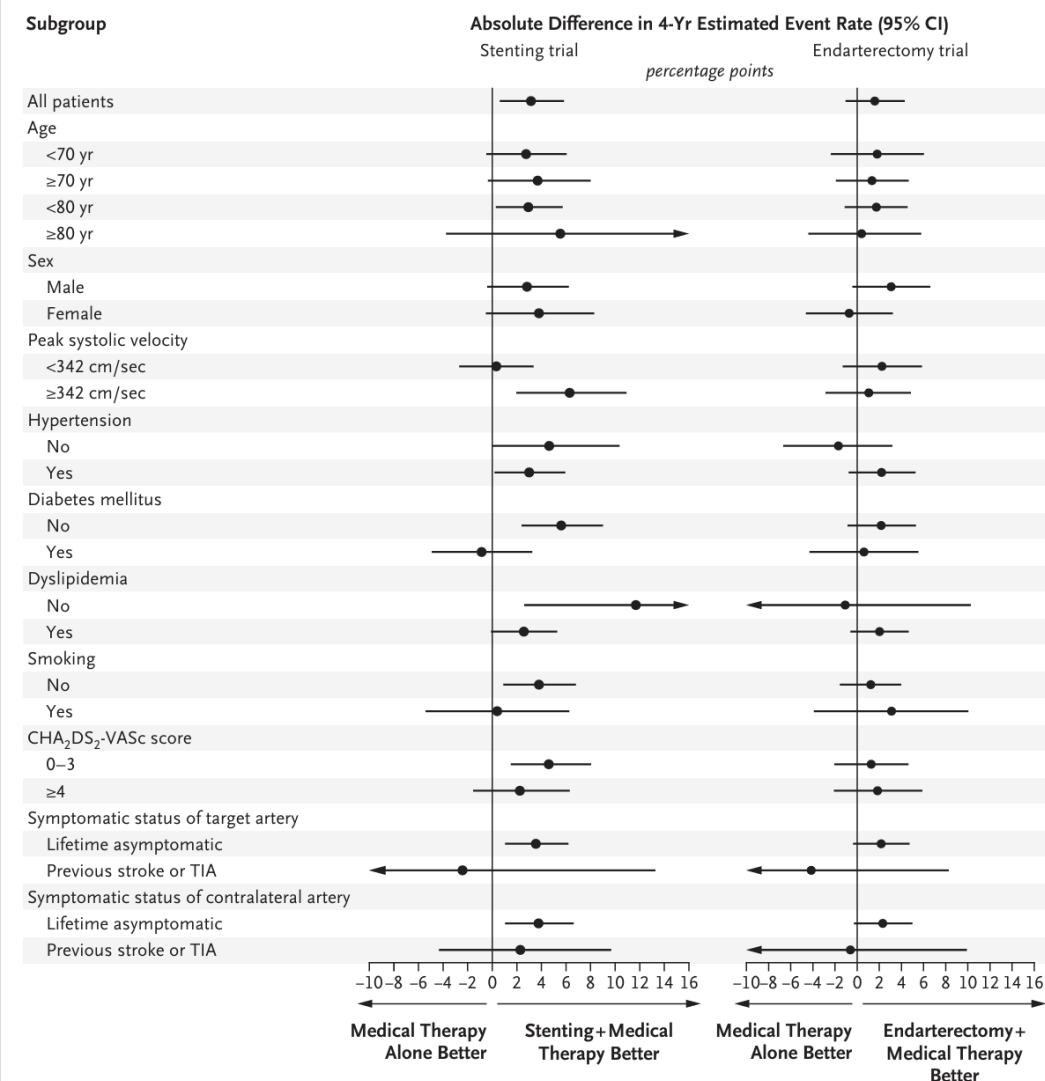


Figure 3. Absolute Between-Group Differences in the 4-Year Estimated Event Rate in the Stenting and Endarterectomy Trials, According to Risk-Factor Subgroups.

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The misleading MRI-infarction row is withheld pending exact secondary-composite/appendix reconciliation. The main report's tissue-based composites include specified death and cerebral infarction/hemorrhage components and cannot be labeled routine MRI lesions.

Tipping Point Analysis



Methodology (supplement): sensitivity analysis - 200 simulated data sets (1000 replications) to determine event count threshold required to shift median p-value across the $\alpha = 0.05$ boundary.

CAS Trial

Events required to lose significance ($P > 0.05$):

Remove events from Medical Arm

≥ 4 Events

Add events to Stenting Arm

≥ 3 Events

CEA Trial

Events required to gain significance ($P < 0.05$):

Add events to Medical Arm

≥ 6 Events

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For CEA, the reported tipping analysis required at least 7 events removed from the CEA arm or at least 6 added to the medical arm to cross the significance boundary. Changing statistical significance does not nullify or reverse an observed treatment effect.

[pmc.ncbi.nlm.nih.gov/articles/PMC2932446/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC2932446/)

[pmc.ncbi.nlm.nih.gov/articles/PMC13148431/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC13148431/)

Conclusions



Temporal Trends

Reduction in revascularization benefit

Natural history → lower event rates → modest CAS benefit

Parallel trends

CAS risks ↓ & improved natural history of asymptomatic high-grade carotid stenosis

Medical treatment could have been better

Potentially eliminating benefit of CAS

Stroke severity

Disabling strokes uncommon in CREST-2



Implications

Risk balance

Does long-term benefit > periprocedural risk?

The operator factor

Outcomes dependent on operator volume & complication rates

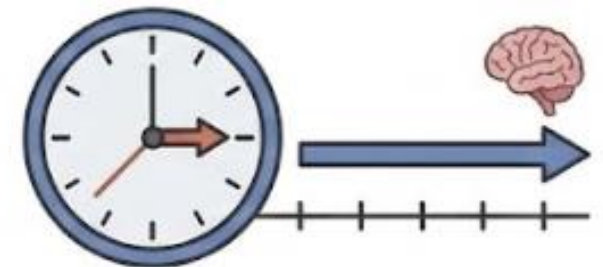
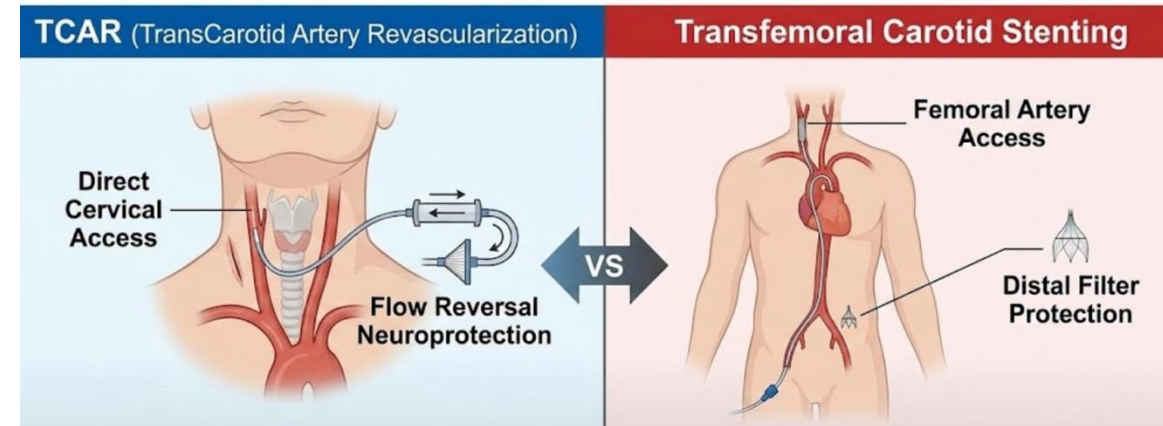
Intensive medical management

We can conclude that there is no longer a role for routine carotid endarterectomy in persons with asymptomatic stenosis.

— BROWN & BONATTI, NEJM EDITORIAL

Limitations

- Open-label
- Medical tx
 - PCSK9i
 - GLP-1RA, SGLT2i
 - BP
- Limited generalizability to low-volume centers
 - real-world complication rates $\uparrow$
- Longer-term Δ in events, subclinical injury, cognition, & stent complications
- TCAR



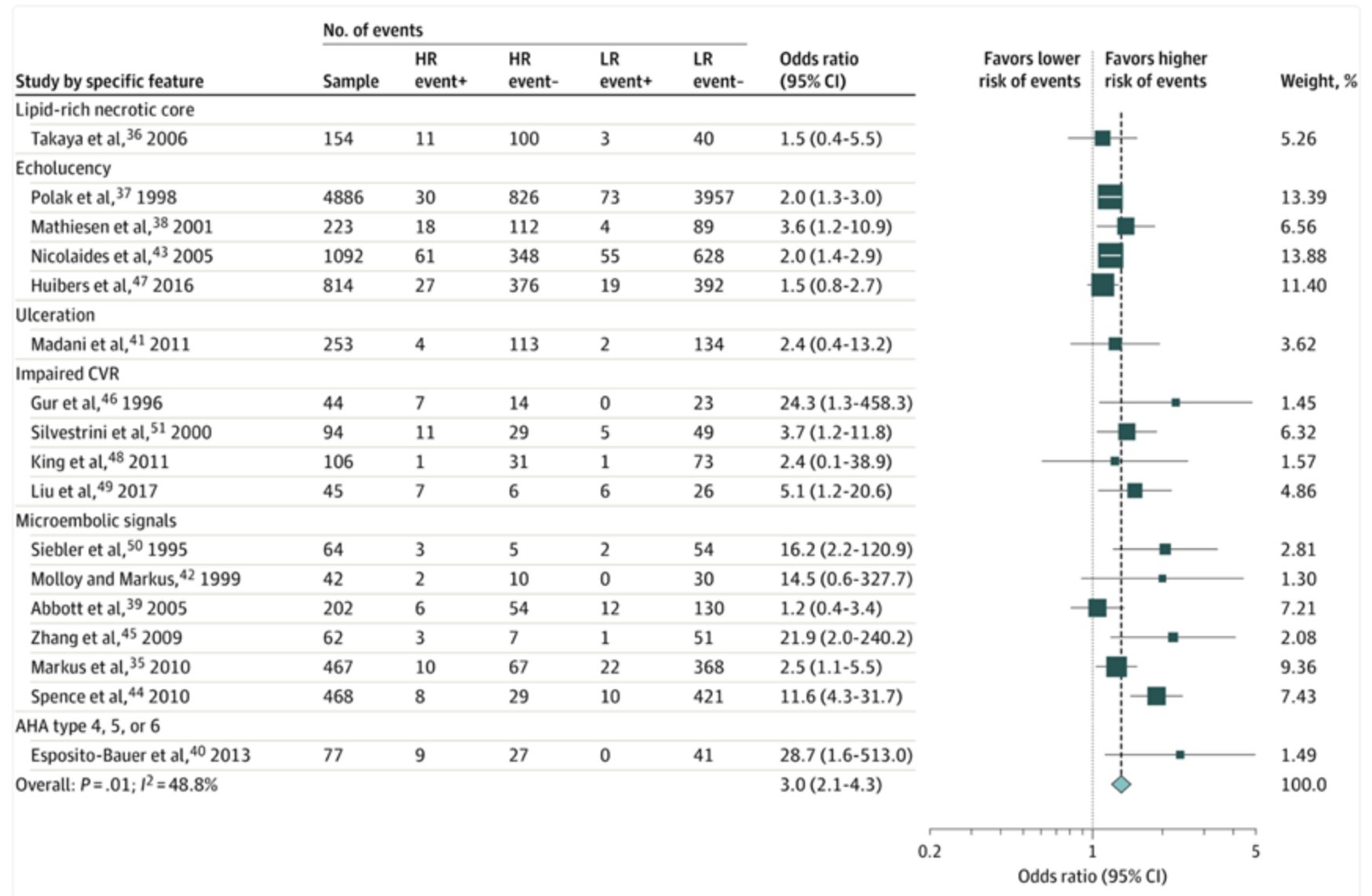
Implementation

Shared decisions

NNT ~31 for the four-year primary composite.

High-risk plaque features are prognostic, not a validated CAS selection rule.

Figure 2. Risk of Ipsilateral Ischemic Cerebrovascular Events in Patients With Asymptomatic Carotid Stenosis With High-risk Features.



JAMA Neurol. 2020 Aug 3;77(12):1–12.

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The approximate NNT of 31 is for the 4-year primary composite, including perioperative death, not stroke alone. High-risk plaque features are prognostic observational findings; they are not a validated rule identifying who specifically benefits from CAS.

pubmed.ncbi.nlm.nih.gov/41269206/

pubmed.ncbi.nlm.nih.gov/32744595/

doi.org/10.1056/NEJMoa2508800

pmc.ncbi.nlm.nih.gov/articles/PMC7400201/

